2.1 $L = [\langle a,g\rangle^3, \langle h,e,g\rangle^3]$

Miss 1	$=1$	$0 = 0$
Cons $1+1+1$	$=3$	$1+1+2+1 = 5$
Prod $1+1+1$	$=3$	$1+1+1+1 = 4$
Rem 1	$=1$	$0 = 0$

$M=3 \quad C=24 \quad P=25 \quad R=3$

$$\text{Fitness}(m) = \frac{1}{2}\left(1-\frac{M}{C}\right) + \frac{1}{2}\left(1-\frac{R}{P}\right) = \frac{1}{2}\left(1-\frac{3}{24}\right) + \frac{1}{2}\left(1-\frac{3}{25}\right) \approx 0.878$$

2.2

① Miss 2	0	$M=20$
Cons $1+1+2+1+2+1=8$	$1+1+1+1+2+1 = 7$	$C=108$
Prod $1+2+1+2+1+1=8$	$1+2+1+1+1+1=7$	$P=108$
Rem 2	0	$R=20$

$$\text{Fitness} = \frac{1}{2}\left(1-\frac{M}{C}\right)+\frac{1}{2}\left(1-\frac{R}{P}\right) = 0.815$$

② $\langle a\rangle$

③ b is the correct alignment

2.3

① Miss 0	1	1	0
Cons $1+1+1+1+2+1=7$	$1+1+2+2+1=7$	$1+1+1+2+1=6$	$1+1+1=3$
Prod $1+1+2+1+1+1=7$	$1+2+1+2+1=7$	$1+1+2+1+1=6$	$1+1+1=3$
Rem 0	1	1	0

$M=7 \quad C=35+21+24+30=110 \quad P=110 \quad R=7$

$$\text{Fitness} = \frac{1}{2}\left(1-\frac{M}{C}\right)+\frac{1}{2}\left(1-\frac{R}{P}\right) = 0.936$$

②

Trace	≫	a	b	d	c	d	e	t
Model	s	a	≫	≫	c	≫	e	t

Cost $=4$ $\quad \text{Fitness} = 1 - \frac{4}{7+2} = 0.55$

2.4

① Miss 2	$=20$	$0 = 0$	$M=20$
Cons $1+1+2+1+1+1$	$=70$	$1+1+1+2+2+1=40$	$C=110$
Prod $1+2+1+1+1+1$	$=70$	$1+2+1+1+2+1=40$	$P=110$
Rem 2	$=20$	$0=0$	$R=20$

$$\text{Fitness} = \frac{1}{2}\left(1-\frac{M}{C}\right)+\frac{1}{2}\left(1-\frac{R}{P}\right) = 1-\frac{20}{110} = \frac{9}{11} = 0.818$$

② $\langle a\rangle$

③ a is the correct one.

2.5

① Miss $1+1$	$=20$	1	$=5$	$M=25$
Cons $1+2+1+1+2$	$=70$	$1+2+1+1+1+1+1$	$=35$	$C=105$
Prod $1+2+1+1+1+1$	$=70$	$1+2+1+2+1+1$	$=40$	$P=110$
Rem 2	$=20$	2	$=10$	$R=30$

$$\text{Fitness} = \frac{1}{2}\left(1-\frac{25}{105}\right)+\frac{1}{2}\left(1-\frac{30}{110}\right) = 0.745$$

② $\langle f\rangle$

③ a is the only correct one

2.6

1.

M	0	$= 0$	1	$= 5$	$M = 5$
C	$1+2+1+1^{\times 1}+2+2+1$	$= 110$	$1+2+1+1+2$	$= 35$	$C = 145$
P	$1+2+2+1+1^{\times 1}+2+1$	$= 110$	$1+2+2+1+1+2$	$= 45$	$P = 155$
R	0	$= 0$	3	$= 15$	$R = 15$

$$\text{Fitness} = \frac{1}{2}\left(1 - \frac{15}{145}\right) + \frac{1}{2}\left(1 - \frac{15}{155}\right) = 0.8999$$

2. <f>

3. C is the only correct one

2.7

In order of fitting C-A-B

2.8

2.9

2.10